INCH-POUND
MIL-H-24299A(SH)
5 August 1994
SUPERSEDING
MIL-H-24299(SHIPS)
12 July 1967
(See 6.8)

MILITARY SPECIFICATION

HEATERS, LUBRICATING OIL AND FUEL, ELECTRIC NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heaters that are stationary, self-contained, and electrically operated. The heaters are used to heat lubricating oil or fuel for purification or warm-up.

1.2 Classification. Heaters are of the following types and styles as specified (see 6.2):

Type A - Immersion heater
Type B - Circulation heater
  Style 1 - Horizontal mount
  Style 2 - Vertical mount

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4540
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
TT-P-645 - Primer, Paint, Zinc-Molybdate, Alkyd Type.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-G-1149 - Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness.
MIL-S-1222 - Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws and Nuts.
MIL-C-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices.
MIL-L-9000 - Lubricating Oil, Shipboard Internal Combustion Engine, High Output Diesel.
MIL-P-15024 - Plates, Tags, and Bands for Identification of Equipment.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-C-15726 - Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).
MIL-L-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.
MIL-S-19622 - Stuffing Tubes, Nylon; and Packing Assemblies; General Specification for.
MIL-F-20042 - Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-H-22577 - Heating Elements, Electrical: Cartridge, Strip and Tubular Type.
MIL-S-22698 - Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural.
MIL-T-23648 - Resistor, Thermal, (Thermally Sensitive Resistor) Insulated, General Specification for.
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0-156.
MIL-T-24388 - Thermocouple and Resistance Temperature Detector Assemblies, General Specification for (Naval Shipboard).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

MIL-P-24691 - Pipe and Tube, Carbon, Alloy and Stainless Steel, Seamless and Welded, General Specification for.
MIL-P-24691/1 - Pipe and Tube, Carbon Steel, Seamless.
MIL-G-24716 - Gaskets, Metallic-Flexible Graphite, Spiral Wound.
MIL-R-83248 - Rubber Fluorocarbon Elastomer, High Temperature, Fluid and Compression Set Resistant.
MIL-R-83248/1 - Rubber, Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant, O-Rings, Class 1, 75 Hardness.
MIL-R-83248/2 - Rubber, Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant, O-Rings, Class 2, 90 Hardness.

STANDARDS

FEDERAL

FED-STD-791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-769 - Thermal Insulation Requirements for Machinery and Piping.
MIL-STD-1399 Section 300 - Interface Standard for Shipboard Systems Section 300 Electric Power, Alternating Current. (Metric)
MS15795 - Washer, Flat-Metal, Round General Purpose.
MS16187 - Stud, Continuous Thread (Bolt Stud), B16 Alloy Steel, High Temperature Service (875-1000°F).
MS17828 - Nut, Self-Locking, Hexagon, Regular-Height, (Non-Metallic Insert) 250°F Nickel-Copper Alloy.
MS17829 - Nut, Self-Locking, Hexagon, Regular Height, 250°F, (Non-Metallic Insert) Non-Corrosion-Resistant Steel.
MS17830 - Nut, Self-Locking, Hexagon, Regular Height, 250°F (Non-Metallic Insert) CRES.
MS18229 - Plug for "O" Ring Gasket.
MS21044 - Nut, Self-Locking, Hexagon, Regular Height, 250°F, 125 KSI FTM and 60 KSI FTM.
MS35307 - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Corrosion Resisting, Passivated, UNC-2A.
MS35311 - Screw, Cap, Hexagon Head-Nickel-Copper Alloy, Plain Finish, UNC-2A.
MS35649 - Nut, Plain, Hexagon, Machine Screw, UNC-2B.
MS51412 - Washer, Flat (Round, Steel, General Purpose).
MS51472 - Nut, Plain, Hexagon, Carbon Steel, UNC-2B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B16.5 - Pipe Flanges and Flanged Fittings.
MC96.1 - Temperature Measurement Thermocouples.

(Application for copies should be addressed to the American National Standards Institute Inc., 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DoD adopted)
A 182 - Standard Specification for Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
A 204 - Standard Specification for Pressure Vessel Plates, Alloy Steel, Molybdenum. (DoD adopted)
A 312 - Standard Specification for Seamless and Welded Austenitic Stainless Steel Pipes.
A 515 - Standard Specification for Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher-Temperature Service. (DoD adopted)
A 516 - Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate and Lower-Temperature Service. (DoD adopted)
D 974 - Standard Test Method for Acid and Base Number by Color-Indicator Titration. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.1.1 Operating fluid. The heater will be used to heat lubricating oil or fuel as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

3.2 Materials. Materials shall be as specified in table I, table II, or table III as specified (see 6.2). In cases where a choice of materials is given in the tables and a specific material is preferred, the acquisition requirements will so indicate.

TABLE I. Ferrous materials.

Part	Material	Applicable document
Shell and pipe connections	Tubing, seamless drawn carbon steel or Plate, welded, carbon-silicon steel, marine boiler quality	MIL-P-24691, MIL-P-24691/1 ASTM A 515 ASTM A 516 ASTM A 204
Head and mounting plate	Plate, carbon-silicon steel, marine boiler quality	ASTM A 515 ASTM A 516 ASTM A 204
Flanges for shell and pipe connections	Steel, carbon	ANSI B16.5
Plates and mounting lug	Plate, carbon-silicon steel, marine boiler quality or Plate, steel, carbon, structural	ASTM A 515 ASTM A 516 ASTM A 204 MIL-S-22698
Stay rods	Steel, carbon	MIL-S-22698
Plugs	Steel, carbon	MS18229
Studs	Steel (grade B16)	MS16187
Screws, bolts	Steel (grade 5 or grade B16)	MIL-S-1222
Nuts	Steel, carbon, zinc coated	MS51472
Nuts, self-locking	Steel, zinc plated	MS17829, MS21044
Washers	Steel, carbon, zinc coated	MS51412

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

TABLE II. Nonferrous materials.

Part	Material	Applicable document
Shell and pipe connections	Copper-nickel alloy, composition 90-10 or Copper-nickel alloy, composition 70-30	MIL-T-16420 MIL-C-15726
Head and mounting plate	Copper-nickel alloy	MIL-C-15726
Flanges for shell and pipe connections 1/	Copper-nickel alloy	MIL-C-15726
Plates and mounting lugs	Copper-nickel alloy	MIL-C-15726
Stay rods	Nickel-copper-aluminum alloy or Copper-nickel alloy, composition 70-30	QQ-N-286 MIL-C-15726
Studs	Nickel-copper or Nickel-copper-aluminum	MIL-S-1222 MIL-S-1222
Screws, bolts	Nickel-copper Nickel-copper-aluminum	MS35311 MIL-S-1222
Nuts Nuts, self-locking	Nickel-copper Nickel-copper	MS35649 MS17828
Washers Plugs	Nickel-copper Nickel-copper alloy or Copper-nickel alloy	MS15795 MS18229

1/ Nonferrous flanges shall meet the dimensional requirements in accordance with MIL-F-20042.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

TABLE III. CRES materials.

Part	Material	Applicable document
Shell and pipe connections	304L or 316L	ASTM A 312
Head and mounting plate	304L or 316L	ASTM A 167
Flanges for shell and heater and pipe connections	304L or 316L	ASTM A 182
Plates and mounting supports	Plate, carbon-silicon steel, marine boiler quality or Plate, steel carbon structural	ASTM A 515 ASTM A 516 ASTM A 204 MIL-S-22698
Stay rods	CRES	ASTM A 312
Plugs	CRES	MS18229
Screws, bolts	CRES CHRES (A286)	MS35307 MIL-S-1222
Nuts	CRES CHRES (A286)	MS35649 MIL-S-1222
Nuts, self-locking	CRES	MS17830

3.2.1 Gaskets and stuffing tubes. The following materials shall be used for gaskets and stuffing tubes, as applicable:

(a) Metallic Flexible Graphite, Spiral Wound in accordance with MIL-G-24716
(b) O-rings in accordance with MIL-R-83248 and MIL-R-83248/1 (MIL-R-83248/2, 90 durometer for threaded bosses).
(c) Synthetic rubber in accordance with MIL-G-1149.
(d) Stuffing tubes, nylon in accordance with MIL-S-19622.

3.2.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.3 Prohibited materials. Asbestos, cadmium, and mercury shall not be used in the manufacture of the heaters or controls. In addition, zinc plating of parts to be immersed in oil is prohibited.

3.3 Construction. Construction shall be as specified in 3.3.1 through 3.3.15.9 (see 6.3).

3.3.1 Heating unit assembly. The heating unit assembly shall consist of a circular mounting flange through which heater elements are inserted and welded to the plate at each penetration. The other end of the elements shall have spacers, or be braced between elements, for rigidity and shall have a means of support to the tank or heater shell. The unit shall be constructed to permit easy removal of connected piping. Mounting flanges shall have threaded breakaway holes. These holes may also be used for inserting lifting eyebolts. All threaded fasteners to be immersed in the heated fluid shall be welded to prevent loosening.

3.3.2 Heater elements. Heater elements shall be type III tubular in accordance with MIL-H-22577 with seamless sheathing and ceramic-to-metal hermetic type end seals. Unless otherwise specified (see 6.2) sheath material shall be as specified in table IV. Heating elements shall be bright annealed after all fabrication steps up to insertion into the mounting plate.

TABLE IV. Sheath material.

Fluid	Type of heater	Sheath material
Lube oil	A B	CRES 304L or 316L Steel
Fuel, service	A or B	CRES 304L or 316L
Fuel, transfer	B	Inconel 625

3.3.3 Terminal box. The electrical connections to the heater elements shall be enclosed in a watertight terminal box mounted on a flange and provided with a removable cover. A connection wiring diagram shall be glued to the underside of the terminal box cover. Holes for cable connections to the terminal box shall be provided as specified (see 6.2). Terminal connections shall be in accordance with MIL-STD-454. Electrical creepage and clearance distances shall be in accordance with MIL-E-917.

3.3.4 Mounting flange. The mounting flange shall be marked to ensure installation of the heating unit assembly in only one position. In order to ensure interchangeability, the tolerance for the location of the drilled flange holes shall be within 0.010 inch of the true design location. The mounting surface shall be finished to ensure a leak-tight seal with the gasket used (see 3.3.14.1(a) and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

3.3.5 Temperature sensing devices. Temperature sensing devices shall be provided with each heater and shall be of the types specified (see 6.2). Unless otherwise specified (see 6.2), the same type of sensor used to input to the controller shall be used to input to alarms, controls and protective devices.

3.3.5.1 Temperature sensing device operation. Temperature sensing device operation shall be verified when heaters are tested for quality conformance as specified in 4.4.

3.3.5.2 Thermostatic switch. Thermostatic operating switches shall be of an adjustable immersed liquid filled bulb type in accordance with MIL-C-2212.

3.3.5.2.1 Thermostatic switch installation shall allow for periodic removal for calibration or replacement without damage.

3.3.5.3 Thermocouple. Thermocouples shall be type K in accordance with MIL-T-24388 or type J in accordance with ANSI MC96.1.

3.3.5.4 Thermistor. When specified (see 6.2), a thermistor temperature detector in accordance with MIL-T-23648 shall be used in conjunction with a solid-state control device.

3.3.5.5 Resistance temperature element. When specified (see 6.2), a resistance temperature element in accordance with MIL-T-24388 shall be used in conjunction with a solid-state control device.

3.3.6 Heater control devices. When specified, the heaters shall be activated by a controller to cycle the heater to maintain a specified liquid temperature range (see 6.2 and appendix). The control circuits shall be arranged so that manual reset will be necessary after operation of the over-temperature safety device, low fluid level switch or electrical overloads. Controllers for heaters that operate for multiple services and/or different temperature settings shall be provided with a positive means to prevent simultaneous operation of more than one mode, such as a selector switch or interlocks.

3.3.6.1 Controller characteristics. Controllers shall have characteristics as specified (see 6.2) as follows:

(a) Solid state, magnetic contactor, and so forth (see appendix).
(b) Power requirements.
(c) Specific degree of heater control required.

3.3.7 Reliability. The heater assembly, temperature control device, and heater control device shall be designed for 10,000 hours of trouble-free operation and 50,000 on-and-off cycles at normal operating conditions.

3.3.8 Finish. Unprotected external surfaces of non-corrosion resistant ferrous parts shall be treated as follows:

(a) Sandblasted clean.
(b) Primed with one coat of primer in accordance with TT-P-645.
(c) Painted with one coat of enamel in accordance with MIL-E-15090.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

Internal surfaces that will come in contact with the heated fluid shall not be painted, plasticized, or zinc coated.

3.3.9 Fasteners. Fasteners shall be identified on drawings by their part or identification numbers (PINs) from the part standards as specified in tables I, II, and III.

3.3.9.1 Commercial utility hardware. Commercial items such as screws, nuts and washers may be used in non-critical applications provided they possess properties similar to and are replaceable by standard parts per 3.3.9.

3.3.10 Welding and allied processes. Welding and allied processes shall be in accordance with MIL-STD-278.

3.3.11 Voltage. The operating voltage shall be either 440 or 115 volts, 60 hertz (Hz); 115 or 230 volts direct current (Vdc); or special voltages in accordance with MIL-STD-1399, section 300, as specified (see 6.2). The element shall operate satisfactorily on voltages up to 105 percent of the specified rating. When heating elements are intended for operation on a voltage other than line voltage (by series or wye connections), the line to line voltage shall also be indicated. The heating element shall be insulated for the line voltage and shall have electrical creepage and clearance dimensions for line voltage in accordance with MIL-E-917. Unless otherwise specified (see 6.2), control devices shall operate on the same voltage as the heating elements.

3.3.12 Safety. Equipment shall be constructed to ensure safety to operating and maintenance personnel in accordance with MIL-E-917.

3.3.13 Over-temperature safety devices.

3.3.13.1 Element temperature. When specified (see 6.2) a temperature sensing device shall be firmly clamped to an inner heating element to measure sheath temperature and shall be set as specified in table V.

TABLE V. Over-temperature safety device set points.

Fluid	Specification	Set point
Lubricating oil, steam turbine (2190TEP)	MIL-L-17331	450 ± 50°F
Lubricating oil, gas turbine	MIL-L-23699	450 ± 50°F
Lubricating oil, diesel (9250)	MIL-L-9000	450 ± 50°F
Fuel, Naval distillate	NATO Code F-44 NATO Code F-76	315 ± 50°F

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

3.3.13.2 Fluid temperature. When specified (see 6.2) a temperature sensor shall be installed to sense the highest local bulk fluid temperature produced by the heater. The sensor shall be internal to a type B heater. The control shall be set to interrupt power to the heater when the fluid temperature rises above the specified thermostat setpoint (see 6.2) and reapply power when fluid temperature drops below setpoint. This control shall operate automatically and shall not require manual reset nor be connected to any alarms.

3.3.14 Type A heaters. Type A heaters shall be as specified in 3.3.14.1 and 3.3.14.2.

3.3.14.1 Heater unit. The heaters shall be of the flanged immersion type with the following characteristics as specified (see 6.2):

(a) Size, type and surface finish of mounting flange.
(b) Power rating.
(c) Immersion length.
(d) Hole diameter for heater insertion.
(e) Weight limitation.

3.3.14.2 Watt density. The watt density for type A heaters shall be not greater than 16 watts per square inch of heating element.

3.3.15 Type B heaters. Type B heaters shall be as specified in 3.3.15.1 through 3.3.15.9.

3.3.15.1 Shell. The heater assembly shall be enclosed in a shell with inlet and outlet piping connections to a lubricating oil or fuel circulation system. Style 1 (horizontal mount) heaters shall have inlet and outlet connections in an upward orientation. Style 2 (vertical mount) heaters shall have the inlet connection at the bottom of the shell and the outlet at the top.

3.3.15.2 Pipe connections. Flanged inlet and outlet connections shall be welded to the shell. Straight threaded connections in accordance with MS16142 shall be provided for draining the heater and for continuous venting. These connections shall be closed by O-ring gasketed plugs in accordance with MS18229. Pipe connections shall be of the size, type and (for flanges) surface finish specified (see 6.2)

3.3.15.3 Heater unit. The heater unit shall be constructed so that the flow of lubricating oil or fuel over all heating surfaces is evenly distributed to prevent localized hot spots. Hot spots for the purposes of this specification are areas where fluid temperatures may exceed outlet temperatures by as much as 40 degrees Fahrenheit (°F). Heating elements shall have an unheated length which extends, as a minimum, from the flange to the centerline of the outlet nozzle.

3.3.15.4 Mounting. Heaters may have mounting lugs welded to the heater shell. The lugs shall be drilled and tapped for bolting the heater to the ship structure. The lugs shall be flush with the insulation and lagging, if provided. Clamp bands or other means of mounting to the ship's structure may also be acceptable. Specific mounting configuration shall be as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

3.3.15.5 Thermal insulation. Heaters shall be provided with a heatshield or they shall be insulated and lagged in accordance with MIL-STD-769 as specified (see 6.2). The lagging shall not interfere with removal of the heater from the shell. Heater lagging shall be jacketed; jacket material shall be CRES in accordance with ASTM A 167, type T304.

3.3.15.6 Watt density. The watt density shall be not greater than 20 watts per square inch of heating element.

3.3.15.7 Capacity. Flow rate, inlet temperature range, outlet temperature, and required minimum temperature rise for the heated fluid shall be as specified (see 6.2).

3.3.15.8 Pressure drop. When operating at maximum rated flow and rated input power and outlet temperature, pressure drop through the heater shall be not greater than 10 pounds per square inch (lb/in^2) unless otherwise specified (see 6.2).

3.3.15.9 Limiting dimensions. Length, width, height, weight, and element withdrawal dimensions shall be as specified (see 6.2).

3.4 Performance characteristics. Performance characteristics shall be as specified in 3.4.1 through 3.4.6.

3.4.1 Operating pressure. Unless otherwise specified (see 6.2), the heater shall operate at 150 lb/in^2 pressure and shall withstand a hydrostatic test of 225 lb/in^2 (see 4.7).

3.4.2 Insulation resistance. Hot and cold insulation resistance tests shall be performed after all welding and allied processes have been completed. The insulation resistance tests as specified in 4.8 shall be in accordance with MIL-H-22577.

3.4.3 Performance for type B heaters. Type B heaters shall perform in accordance with specification requirements when tested as specified in 4.9. This test shall obtain information for different flow rates and inlet temperatures.

3.4.4 Vibration. Heaters shall withstand the environmental vibration test specified in 4.10 without mechanical damage or malfunctioning.

3.4.5 Shock. Heaters shall withstand high-impact shock as specified in 4.11 for grade A class I equipment in accordance with MIL-S-901.

3.4.6 Dielectric strength. Heating elements shall withstand the dielectric test specified in 4.12.

3.5 Identification and information plates. Each heater shall have an identification plate in accordance with types A, B, or C of MIL-P-15024, mounted on the terminal box or on the heater lagging. The plate shall contain the following information:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

(a) Fluid to be heated.
(b) Capacity in gallons per minute (type B only).
(c) Serial number and type of heater.
(d) Electrical characteristics: voltage, wattage, number of phases, frequency, and number of elements.
(e) National Stock Number (if available).
(f) Name and address of manufacturer.
(g) Manufacturer's part number.

3.6 Workmanship. Heaters shall be free from defects affecting their appearance or that may affect their operation. Mating surfaces of flanges and mounting pads shall be machined true to provide interchangeability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements. However, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.2.1 Inspection conditions. Unless otherwise specified (see 6.2) all inspections shall be performed in accordance with the test conditions of MIL-STD-202.

4.3 First article inspection. First article inspection shall be performed on one heater of each type and capacity offered for delivery on the contract or order. The heater shall be subjected to the first article tests specified in table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

TABLE VI. Examination and tests.

Inspection	Requirement	Test method	First article	Quality conformance
Examination				
Material & Construction	3.2 and 3.3	4.5	X	X
Calibration Temp-sensors Over-temperature Safety devices	3.3.5.1 and 3.3.13	4.6	X	X
Hydrostatic test	3.4.1	4.7	X	X
Insulation resist.	3.4.2	4.8	X	X
Performance (type B)	3.4.3	4.9	X	
Vibration	3.4.4	4.10	X	
Shock	3.4.5	4.11	X	
Dielectric	3.4.6	4.12	X	X

4.3.1 Manufacturer's production inspection. If production inspection data reveal variations beyond the normal manufacturing tolerance, a complete first article inspection shall be made on that particular type or size of heater to demonstrate that it conforms to this specification.

4.4 Quality conformance inspection. Each heater shall be subjected to the quality conformance inspections specified in table VI (see 6.3).

4.5 Examination. Each heater offered for delivery shall be examined to ascertain that it is in accordance with this specification. Any heater in the sample containing one or more defects shall not be offered for delivery (see table VII).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

TABLE VII. Classification of defects.

Categories	Defects
Critical:	None defined.
Major:	
101	Size not as specified.
102	Incomplete, component parts missing (heating elements, over-temperature safety, temperature control, float switch as applicable).
103	Materials defective or not as specified.
104	Limiting dimensions exceeded.
105	Flange and mounting holes not as specified.
106	Welding incomplete, not free of cracks, nonfusion, heavy porosity, heavy undercut, slag inclusions.
107	Corner of flanges not smooth, square or grooved (not pressure tight).
108	Access cover plates missing.
109	Gaskets, missing or defective.
110	Splices in wire.
111	Electrical connections, not at terminal of component or terminal blocks.
112	Wiring not neat.
113	Bolts, nuts, and screws not tight, missing, or not as specified.
114	Sharp edges and burrs not removed from parts.
115	Drawings not followed, or wiring diagram missing.
116	Finish not conforming as specified.
117	Marking, identification plate not complete, missing, not permanent, illegible, or not as specified.
118	Creepage and clearance distances not as specified.

4.6 Calibration. The calibration of the temperature control devices and the over-temperature safety devices shall be checked by a controlled temperature liquid bath system or by use of a test thermocouple attached to the sensing bulb or thermowell of the sensing device to indicate the temperature at which the device functions.

4.7 Hydrostatic test. The heater shall be tested at the specified hydrostatic pressure (see 3.4.1 and 6.2).

4.8 Insulation resistance test. This resistance test shall be as specified in 4.8.1 and 4.8.2 (see 3.4.2).

4.8.1 Insulation resistance, cold. The insulation resistance shall be measured with 500 Vdc at room temperature to determine conformance to 3.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

4.8.2 Insulation resistance, hot. The insulation resistance shall be measured with a minimum of 500 Vdc within 5 seconds after operation of the element at maximum operating temperature as specified (see 6.2) to determine conformance to 3.4.2. To prevent the element from being cooled before the test, the ambient fluid shall not be circulated after the power is removed.

4.9 Performance test for type B heater. Type B heaters and associated controls shall be tested as specified in 4.9.1 through 4.9.4.

4.9.1 Capacity test. With inlet fluid temperatures and flow rates as specified in table VIII, the following shall be determined:

(a) Outlet temperature.
(b) Volts and amperes.
(c) Pressure drop from inlet to outlet of heater.
(d) Number and location of heater elements energized.
(e) Thermocouple readings (thermocouples shall be installed to indicate temperatures in areas likely to be hot spots within the heater caused by insufficient flow).

All tests shall be conducted using lubricating oil in accordance with MIL-L-17331 (Military symbol 2190TEP).

4.9.2 Pressure drop. Determine pressure drop from heater inlet to outlet for the flow rates specified in table VIII with the heater operating at rated input power and the specified minimum inlet fluid temperature.

TABLE VIII. Temperatures and flow rates.

Flow (gallons per minute)	25 percent capacity	50 percent capacity	75 percent capacity	100 percent capacity
Inlet temperatures (°F)	40 60 90	40 60 90	40 60 90	40 60 90

4.9.3 Venting. When specified (see 6.2), the ability of the heater to be drained, filled and manually vented shall be demonstrated. The demonstration shall include inclined operation as specified in MIL-E-917 (i.e., 15 degrees in any direction from normal operating position for surface ship heaters and 30 degrees for submarine heaters). Each run shall be for a 1-hour period at rated flow with inlet oil at 90°F temperature and 20 lb/in^2 gage pressure. During the test, air shall be introduced into the inlet. The vent connection shall be opened periodically and checked for presence of air to ensure that the heater is venting properly. At the completion of the test, the elements shall be visually examined for damage and traces of carbon deposits possibly caused by the existence of an unvented air pocket.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

4.9.4 Temperature control (lubricating oil only). With the type B heater completely filled with oil and not circulating, full power shall be applied for 8 hours. The thermostatic switch shall be set to cut in at 155°F, decreasing temperature and set to cut out at 160°F, increasing temperature. Areas of possible hot spots shall be monitored and temperatures recorded at 1-hour intervals. At the conclusion of the test, a sample of oil shall be subjected to the following tests:

(a) Precipitation number (maximum 0.01) in accordance with FED-STD-791.
(b) Neutralization number in accordance with ASTM D 974.

4.10 Vibration test. The heater shall be subjected to the type I vibration test in accordance with MIL-STD-167-1 (see 6.3).

4.11 Shock test. The heater shall be subjected to the type A, high-impact (H.I.) shock test in accordance with MIL-S-901 for grade A, class I equipment (see 6.3).

4.12 Dielectric test. Dielectric test shall be as specified in 4.12.1 through 4.12.3.

4.12.1 General. The dielectric test shall be made after all other tests have been completed. The test voltage shall be equal to twice the rated value of the terminal voltage plus 1000 volts. The frequency shall be not less than 60 Hz, and the waveshape shall approximate a sine wave. The test voltage shall be applied continuously for a period of not less than 1 minute.

4.12.2 Measurements of test voltage. The measurements of the voltage used in dielectric test shall be made by means of a voltmeter that shall derive its voltage directly from the high voltage circuit, either by means of a voltmeter coil placed in the testing transformer or through an auxiliary potential transformer.

4.12.3 Points of application. The test voltage shall be successively applied between each electric circuit and the frame with all other electric circuits and metal parts grounded.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging requirements. Heaters and accessories shall be preserved level A, C, or commercial, packed level A, B, C, or commercial as specified (see 6.2) and marked in accordance with MIL-E-17555 and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2). In addition, for Navy acquisitions, the following applies:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

(a) Navy shipboard stowage fire-retardant requirements.

(1) Treated lumber and plywood. When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material in accordance with MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(2) Fiberboard. When specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements in accordance with PPP-F-320 and amendments thereto.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Types A and B heaters are intended for the following applications:

(a) Type A heaters. Type A heaters are intended for installation in machinery lubricating oil sump tanks to maintain the proper temperature and viscosity of the oil where little circulation or heat is provided (for example, reduction gear sumps).
(b) Type B heaters. Type B heaters are intended for use in lubricating oil and fuel systems where heat is required in purifying or to raise the temperature to operating limits required for the applicable machinery.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type and style of heater (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Whether first article inspection is required (see 3.1).
(e) Operating fluid (see 3.1.1).
(f) Materials (see 3.2)
(g) Sheath material for heater elements (see 3.3.2).
(h) Terminal box hole size and location (see 3.3.3).
(i) Temperature sensing device (see 3.3.5).
(j) Heater control device and temperature range (see 3.3.6 and appendix).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

(k) Controller characteristics (see 3.3.6.1).
(l) Voltage requirements (see 3.3.11).
(m) Over temp safety devices (see 3.3.13.1 and 3.3.13.2).
(n) Type A heater characteristics (see 3.3.14.1).
(o) Pipe connections (see 3.3.15.2).
(p) Mounting configuration (see 3.3.15.4).
(q) Insulation or heat shield (see 3.3.15.5).
(r) Capacity (see 3.3.15.7).
(s) Pressure drop (see 3.3.15.8).
(t) Limiting dimensions (see 3.3.15.9).
(u) Operating pressure if other than as specified (see 3.4.1).
(v) Inspection conditions (see 4.1 and 4.2.1).
(w) Maximum operating temperature (see 4.8.2).
(x) Levels of packaging, bar codes, and options (see 5.1).
(y) Fire-retardant materials (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
3.3	DI-DRPR-81000	Product drawings and associated lists	----
4.4 and 4.9	DI-NDTI-80809	Test/inspection reports	----
4.10	UDI-T-23762	Report, vibration testing	----
4.11	DI-ENVR-80708	Shock test report	----

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5.1 Design changes. A first article inspection must be performed when any change in design affects performance characteristics.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Subject term (key word) listing.

Controller, solid state
Ferrous
Heater, circulation
Heater, immersion
Nonferrous
Safety device, over-temperature
Thermistor

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4540-N001)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

APPENDIX A

CONTROLS FOR HEATERS

10. SCOPE

10.1 This appendix provides information with regard to the controls/ controllers and associated components normally supplied with lubricating oil and fuel heaters covered by this specification. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

10.2 Type A heaters. Applications using type A heaters generally require a heat-up time of approximately 1 hour during which the heaters are energized 100 percent of the time. Immersion heaters are generally installed in tanks. When the heater is on, heat transfer in tanks occurs by conduction and convection at a rate of rise of about one degree per minute. Controllers for immersion heaters are generally of the magnetic type as specified in 30.3. Controllers may be designed to heat up to a specified temperature and shut off, or they may control around a specific temperature.

10.3 Type B heaters. Applications using type B heaters generally require 50 to 120°F temperature rise in the period of time (5 seconds or less) that it takes for one pass through the heater. If the outlet of the heater flows into a tank, control of the tank temperature is more appropriate than control of the heater outlet temperature. Control of tank temperature may require magnetic controllers (see 30.3) or stepper controllers (see 30.4). If the outlet of the heater does not flow into a tank, solid-state controllers (see 30.5) and proportional integral derivative (PID) control (see 30.6) may be required to maintain outlet temperature.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

30. REQUIREMENTS

30.1 All heater controllers shall meet the requirements of MIL-STD-461.

30.2 The requirements of MIL-C-2212 apply to all controllers where certain components such as contactors, pilot lights, pushbuttons and relays, and so forth are included in the design.

30.3 Magnetic contactor. When specified (see 6.2), a magnetic contactor shall be furnished with each heater. The contactor shall be in accordance with MIL-C-2212. The heater control device shall function to de-energize the heater within plus or minus 10°F of the adjusted temperature setting. At the time the control device functions, the maximum temperature of the fluid in any part of the heater shall not exceed the adjusted temperature setting by greater than 15°F. The heater shall be re-energized before the temperature has dropped to 90 percent of the adjusted temperature setting.

30.4 Stepper control. When specified (see 6.2), a circular stepper power control shall be furnished for each heater. Stepper power control shall be by a magnetic across-the-line type controller with the heater load divided into equal parts to be sequentially energized via contactors with a 10-second delay between each sequence and de-energized in a first-on-first-off order. Sequencing control shall be by solid-state control.

30.5 Solid-state controller. When specified (see 6.2), a solid-state controller shall be furnished. The heater control device in this instance shall de-energize the heater within plus or minus 5°F of the adjusted temperature setting. At the time the control device functions, the maximum temperature of the fluid in any part of the heater shall not exceed the adjusted temperature setting by more than 8°F. The heater shall be re-energized before the temperature of the fluid has dropped to 95 percent of the adjusted temperature setting. The controller shall cycle the heating elements by a solid-state thyristor power controller to maintain specified heater outlet temperature. The solid-state thyristor control shall provide enforced zero firing power control. Heater elements shall be divided into three or more circuits having equal KW ratings. Under steady-state conditions, individual circuits shall be brought from 0 to 100 percent of the rating sequentially with increasing demand for heat. The cycle period for thyristor control shall be field adjustable. Thyristors shall be protected by fuses of the type suitable for protecting semi-conductor devices. Thyristor heat sinks shall be suitable for ambient temperatures encountered. The controllers shall shut down the heater upon loss of any thermocouple signal.

30.6 PID controller. When specified (see 6.2), a PID controller shall be furnished with each heater. Heater control temperature shall be sensed by thermocouple. Heater controllers shall be of the magnetic controller/solid-state thyristor type. The controller shall be in accordance with MIL-C-2212 and the controller enclosure shall contain the solid-state PID temperature controller, element over-temperature control, magnetic contactors for shorted SCR, and a solid-state (SCR) thyristor device for proportioning power to heating elements and control transformers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-24299A(SH)	2. DOCUMENT DATE (YYMMDD) 5 August 1994

3. DOCUMENT TITLE
HEATERS, LUBRICATING OIL AND FUEL, ELECTRIC NAVAL SHIPBOARD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
TECHNICAL POINT OF CONTACT: Tom Constallono,03E23	703- 602-7440	332-7440
c. ADDRESS (Include Zip Code) Commander SEA 03R42 2531 Jefferson Davis Hwy Arlington, VA 22241-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:56Z
Check the source to verify that this is the current version before use.